

The four RL components, strictly PS-grounded

1. The hidden state: $E[N, T]$

The PS gives this directly:

"The status of environment for each frequency band at each time step can be recorded as a transmission or a non-transmission."

That sentence is the environment's data structure, a binary matrix of shape $(N_bands \times T_steps)$. The environment holds this internally.

The agent never sees it directly. This is what makes the problem a POMDP rather than a fully-observable MDP.

What populates E comes from the two emitter types the PS names explicitly: **frequency-agile emitters**, whose active band changes over time (so their contribution to E moves horizontally across the grid), and **spatially-scanning emitters**, whose contribution appears only during their illumination windows (so their row in E has periodic gaps).

No other emitter types were invented, those are the only two the PS mentions as the hard cases.

2. The observation space: shape ($N \times 3 + 1$,)

The PS says the receiver operates in "*the absence of prior reliable intelligence*." So the agent cannot be given any pre-loaded knowledge of who is where. **The only information it legitimately has is what it has accumulated itself through its own scan history: hit rate per band, how often each band has been visited, and how stale its knowledge of each band is.**

These **three quantities map** directly to three PS figures of merit:

- Probability of detection
- Average intercept rate
- Average intercept time error, respectively.

The observation is built from these, not from the hidden truth.

3. The action space: Discrete(N)

The PS: "*adjusting receiver's frequency at correct time.*" The agent controls the frequency dimension, which band to listen to. Time advances automatically. So the action is a single integer from 0 to $N-1$. Nothing more. There is no action for dwell duration, no action for gain setting, no action for sensitivity, the PS does not give those to the scheduler.

4. The reward function (hits, misses, and opportunity cost)

The PS gives the training signal explicitly: *"trained based on hits and misses."* Hit = +1, miss = 0. The PS also states the primary objective is *"minimize intercept time and ensure a high interception rate"*

and separately warns that open-loop strategies *"may lose time to nonthreatening emitters by not giving time to new or threatening ones."* That warning is the direct justification for the small -0.1 penalty on immediate revisits with no hit, it is the computational translation of the PS's own critique of wasteful dwell.

+ What the smoke test reveals

The round-robin baseline achieved only **2.2% interception rate** on this 4-emitter scene. That's expected and correct, a fixed sweep cycle completes one full pass every 36 steps, while frequency-agile emitters hop every 3-4 steps and spatially-scanning emitters illuminate for only 1-2 steps per 10-15 step period. The two timescales are completely mismatched, which is exactly the problem the PS is asking you to solve.

That 2.2% is your baseline. A trained RL scheduler operating on the same E_truth grid should achieve a significantly higher interception rate by learning to anticipate the hop patterns and illumination windows from accumulated hit/miss history alone, which is exactly what the PS means by *"trained based on hits and misses."*